

Quantum Mechanics

Lecture 7: Global Phase, Polarization Ellipses, and Unitary Evolution

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Chapter 1

Global Phase, Polarization Ellipses, and Unitary Evolution

1.1 Completing the Polarization Observable

We begin with the unfinished exercise from the preceding lecture. The horizontal and vertical polarization states are

$$|x\rangle = \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \quad |y\rangle = \begin{pmatrix} 0 \\ 1 \end{pmatrix}, \quad (1.1)$$

and the observable that records +1 for horizontal transmission and -1 for vertical transmission is

$$\hat{P}_0 = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}. \quad (1.2)$$

At 45°, the two eigenstates

$$|d_{\pm}\rangle = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ \pm 1 \end{pmatrix} \quad (1.3)$$

belong to

$$\hat{P}_{\pi/4} = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}. \quad (1.4)$$

The homework was to construct the observable for the circular basis

$$|c_+\rangle = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ i \end{pmatrix}, \quad |c_-\rangle = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ -i \end{pmatrix}. \quad (1.5)$$

The spectral rule gives the answer immediately:

$$\hat{P}_c = |c_+\rangle \langle c_+| - |c_-\rangle \langle c_-| \quad (1.6)$$

$$= \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}. \quad (1.7)$$

Indeed,

$$\hat{P}_c |c_{\pm}\rangle = \pm |c_{\pm}\rangle. \quad (1.8)$$

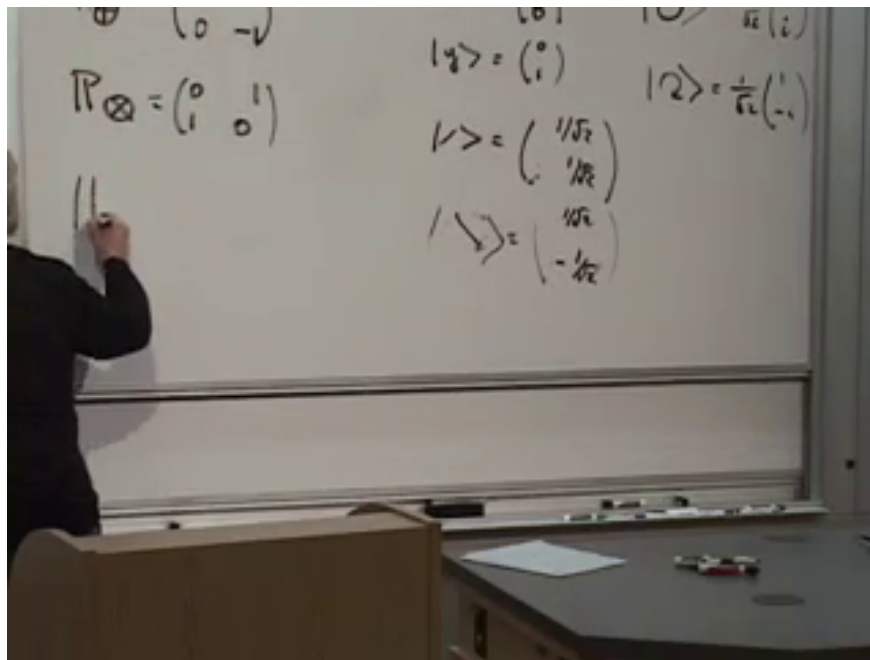


Figure 1.1: The three polarization bases and the circular observable at the end of the homework review.

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The only delicate point in the check is $i^2 = -1$, together with complex conjugation when a ket becomes a bra.

Orthogonality has an operational meaning throughout this discussion. Two orthogonal polarization states can be distinguished with certainty in one ideal analyzer: an apparatus that transmits one rejects the other. This is why the inner product is not decorative notation. It records a measurable relation between preparations.

1.2 What States and Observables Are For

It is useful to pause and compare our present position with the opening classical discussion. For a coin or die, the state is one member of a finite set. A classical observable is simply a numerical function on that set. We might assign $+1$ to heads and -1 to tails, while the die arrives with the labels $1, \dots, 6$. Relabelling the outcomes changes the numerical convention, not the underlying states.

The same idea survives in continuous classical mechanics. A state is a point (q, p) in phase space, and an observable is a function

$$F = F(q, p). \quad (1.9)$$

Position q , momentum p , and $p + q$ are elementary examples. As the phase-space point moves, the numerical value of each function changes.

Quantum mechanics replaces the set of states by a complex vector space and the numerical functions by Hermitian operators. Before discussing motion in that space, however, we should isolate the two principal questions that a state vector already answers.

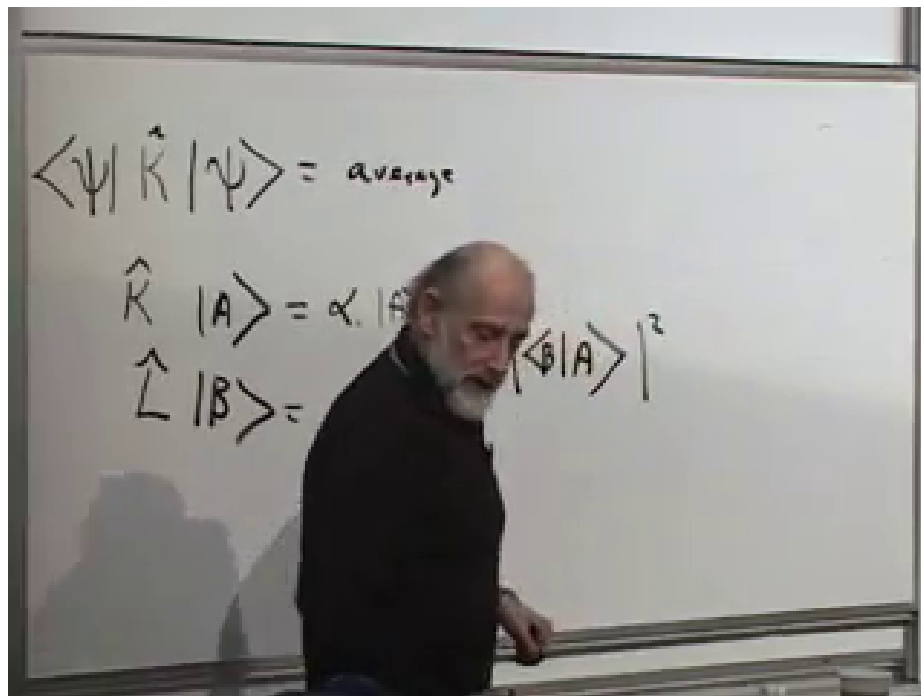


Figure 1.2: Expectation value, eigenstate preparations, and the squared overlap as the two physical uses of a state vector.

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First, for a normalized state $|\psi\rangle$ and observable $\hat{K}$, the ensemble average is

$$\langle \hat{K} \rangle_{\psi} = \langle \psi | \hat{K} | \psi \rangle. \quad (1.10)$$

Prepare the same state repeatedly, measure K on each member of the ensemble, and the empirical mean approaches this number.

Second, suppose

$$\hat{K} |a\rangle = \alpha |a\rangle, \quad \hat{L} |b\rangle = \beta |b\rangle. \quad (1.11)$$

If the preparation makes $K = \alpha$ definite and the subsequent measurement asks whether $L = \beta$, then

$$\text{Prob}(b|a) = |\langle b | a \rangle|^2 = \langle a | b \rangle \langle b | a \rangle. \quad (1.12)$$

Reversing the preparation and test gives the same numerical probability, because $|\langle a | b \rangle|^2 = |\langle b | a \rangle|^2$.

Question from the audience. Does the order of the two overlaps in the probability matter?^a

Response. The two amplitudes are complex conjugates: $\langle a | b \rangle = \langle b | a \rangle^*$. Their product is therefore the same nonnegative real number in either order. The correction on the board changes the amplitude, but not the transition probability.

^aLecture timestamp: 00:14:54–00:15:18.

Average values and transition probabilities both refer to repeated experiments. A single trial returns one allowed outcome; the probability and the average emerge from an ensemble. This common structure leads directly to the question of phase.

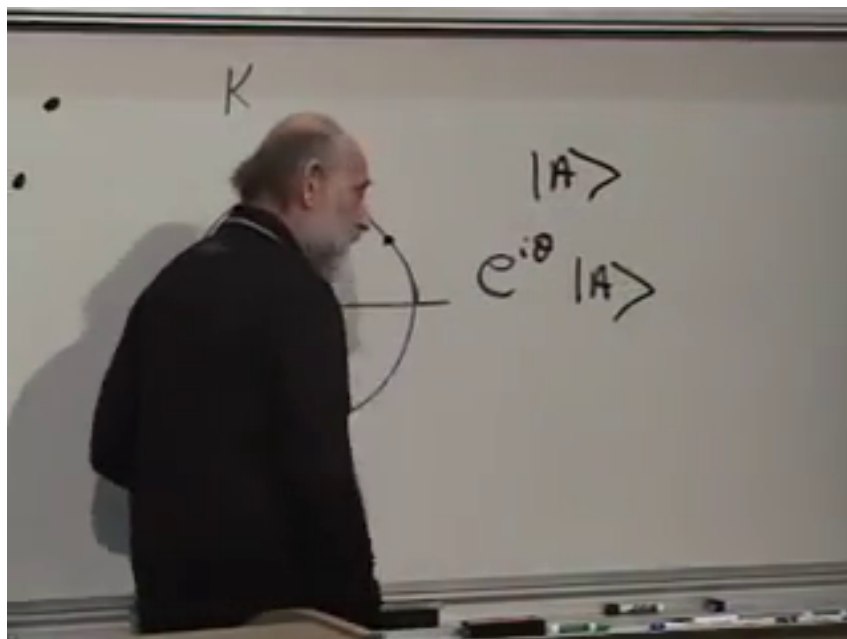


Figure 1.3: The unit circle beside the comparison of $|A\rangle$ and $e^{i\chi}|A\rangle$.

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1.3 A Physical State Is a Ray

1.3.1 Pure Phases

A physical state vector is normalized:

$$\langle\psi|\psi\rangle = 1. \quad (1.13)$$

If $|\psi\rangle = \sum_n c_n |n\rangle$ in an orthonormal basis, this is $\sum_n |c_n|^2 = 1$, the statement that the total probability is one.

Normalization does not remove every redundancy. To see what remains, write a complex number in polar form,

$$z = x + iy = re^{i\chi} = r(\cos \chi + i \sin \chi). \quad (1.14)$$

Multiplication multiplies moduli and adds angles:

$$r_1 e^{i\chi_1} r_2 e^{i\chi_2} = r_1 r_2 e^{i(\chi_1 + \chi_2)}. \quad (1.15)$$

Complex conjugation reflects the number across the real axis,

$$z^* = r e^{-i\chi}. \quad (1.16)$$

A complex number of unit modulus is a *pure phase*:

$$e^{i\chi}, \quad e^{i\chi} e^{-i\chi} = 1. \quad (1.17)$$

The word phase may refer either to the angle χ or to the unit complex number $e^{i\chi}$; the context tells us which is intended.

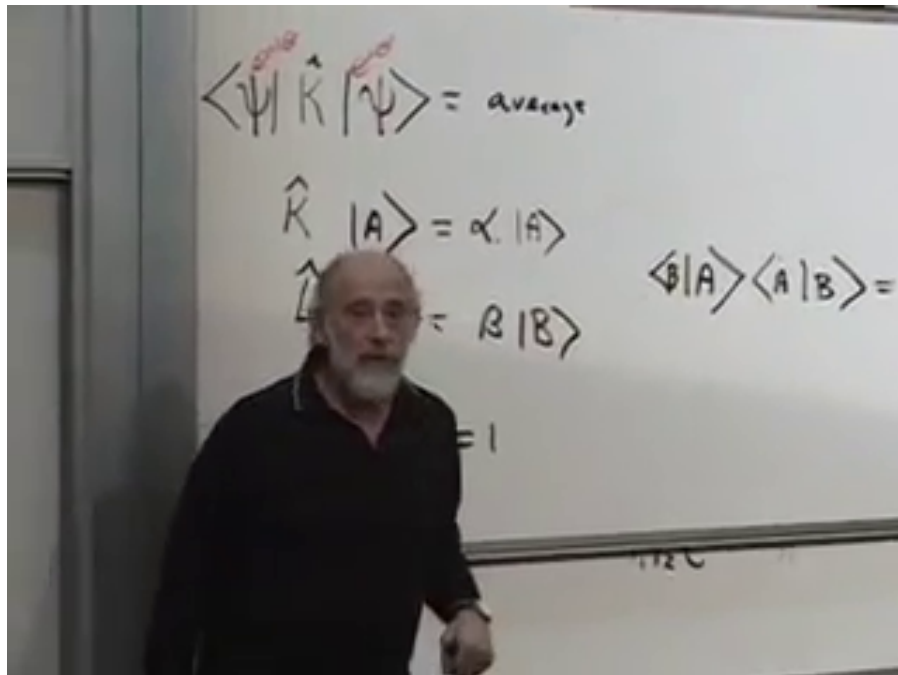


Figure 1.4: The two phase factors cancel in both the operator sandwich and the squared overlap.

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1.3.2 Why the Overall Phase Cancels

Mathematically, $|\psi\rangle$ and $e^{i\chi}|\psi\rangle$ are different vectors. Physically, they represent the same pure state. Let

$$|\psi'\rangle = e^{i\chi}|\psi\rangle. \quad (1.18)$$

The corresponding bra is

$$\langle\psi'| = e^{-i\chi}\langle\psi|. \quad (1.19)$$

Every expectation value is unchanged:

$$\langle\psi'|\hat{K}|\psi'\rangle = e^{-i\chi}\langle\psi|\hat{K}e^{i\chi}|\psi\rangle \quad (1.20)$$

$$= \langle\psi|\hat{K}|\psi\rangle. \quad (1.21)$$

The phase is an ordinary scalar and commutes through $\hat{K}$.

The transition probability is equally insensitive. If $|b'\rangle = e^{i\eta}|b\rangle$, then

$$|\langle b'|a\rangle|^2 = e^{-i\eta}\langle b|a\rangle e^{i\eta}\langle a|b\rangle \quad (1.22)$$

$$= |\langle b|a\rangle|^2. \quad (1.23)$$

Independent phase choices for either state disappear from the absolute square.

The physical object is therefore not one normalized vector but the equivalence class

$$|\psi\rangle \sim e^{i\chi}|\psi\rangle. \quad (1.24)$$

Such an equivalence class is called a *ray* in Hilbert space. In particular,

$$\begin{pmatrix} 0 \\ 1 \end{pmatrix} \sim \begin{pmatrix} 0 \\ -1 \end{pmatrix}, \quad (1.25)$$

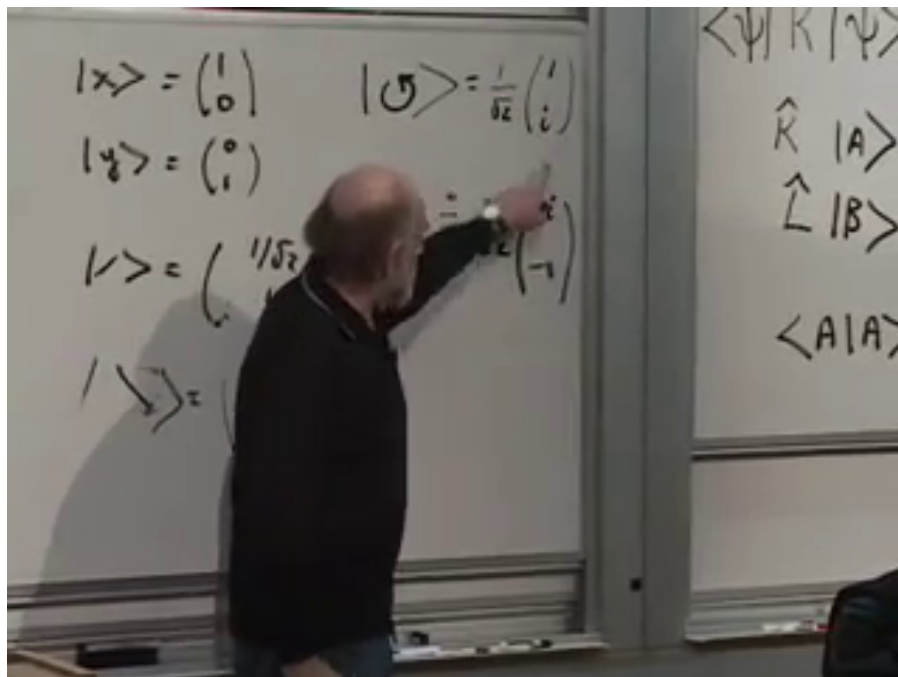


Figure 1.5: Multiplying both circular-polarization components by i changes the vector but not the physical ray.

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because $-1 = e^{i\pi}$. Likewise,

$$\frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ i \end{pmatrix} \sim \frac{1}{\sqrt{2}} \begin{pmatrix} i \\ -1 \end{pmatrix}, \quad (1.26)$$

because the second vector is i times the first.

Question from the audience. What notation should express that two vectors have the same physical significance without saying that they are equal? ^a

Response. Ordinary equality is too strong, because the vectors really differ. The standard statement is that they represent the same ray, written here as $|\psi\rangle \sim e^{i\chi} |\psi\rangle$. What is preserved is every expectation value and transition probability.

^aLecture timestamp: 00:25:49–00:27:50.

1.4 Two Parameters Describe Polarization

Begin with the general two-component vector

$$|\psi\rangle = \begin{pmatrix} \alpha \\ \beta \end{pmatrix}, \quad \alpha, \beta \in \mathbb{C}. \quad (1.27)$$

The two complex entries contain four real parameters. Normalization removes one:

$$|\alpha|^2 + |\beta|^2 = 1. \quad (1.28)$$

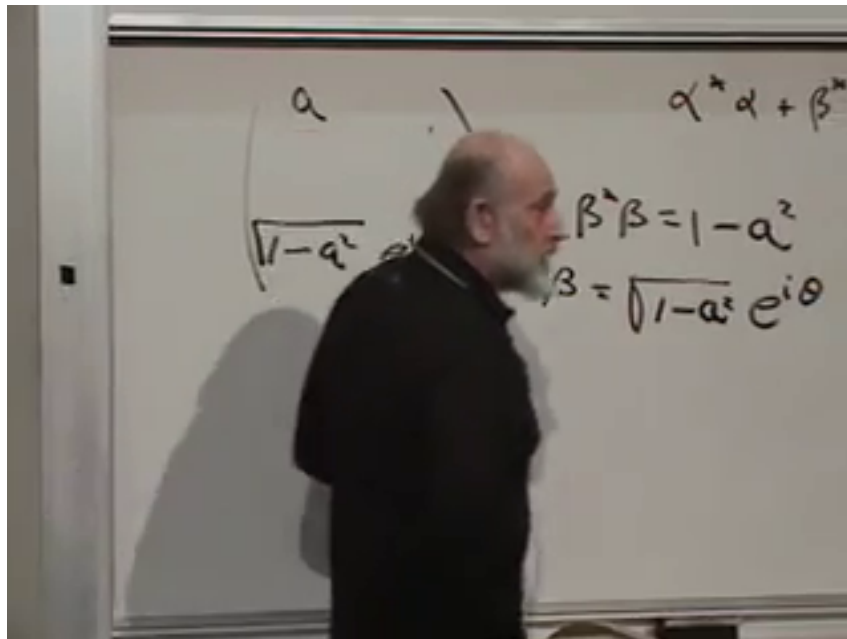


Figure 1.6: Normalization and phase fixing reduce two complex amplitudes to two physical real parameters.

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The irrelevant common phase removes another. Except at the coordinate pole $\alpha = 0$, choose that phase so that $\alpha = a$ is real and nonnegative. Then

$$\beta = \sqrt{1 - a^2} e^{i\phi}, \quad 0 \leq a \leq 1, \quad (1.29)$$

and

$$|\psi\rangle = \begin{pmatrix} a \\ \sqrt{1 - a^2} e^{i\phi} \end{pmatrix}. \quad (1.30)$$

Exactly two real parameters remain: a and the relative phase ϕ .

An equivalent parameterization makes the geometry especially transparent:

$$|\psi\rangle = \begin{pmatrix} \cos(\vartheta/2) \\ e^{i\phi} \sin(\vartheta/2) \end{pmatrix}, \quad 0 \leq \vartheta \leq \pi. \quad (1.31)$$

The rays of a two-state system form a sphere, often called the Bloch sphere or, for polarization, the Poincare sphere. The lecture does not require that name, but its two-parameter count is precisely this geometry.

1.5 From the State Vector to a Polarization Ellipse

For a monochromatic wave moving along z , write the transverse electric field as

$$\mathbf{E}(z, t) = \text{Re} \left[E_0 (\alpha \hat{\mathbf{x}} + \beta \hat{\mathbf{y}}) e^{i(kz - \omega t)} \right]. \quad (1.32)$$

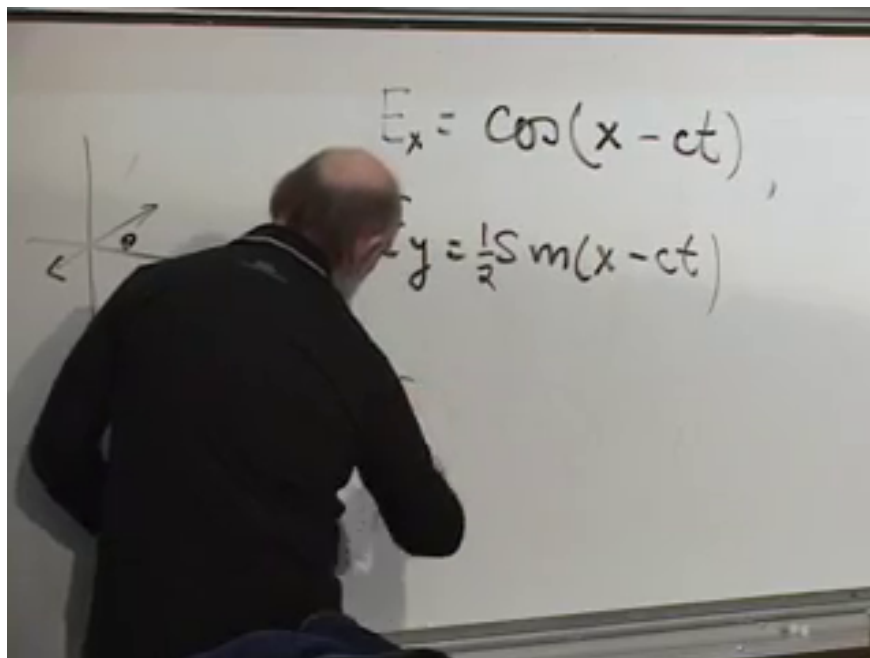


Figure 1.7: Equal quadrature amplitudes give a circle; reducing one amplitude gives an ellipse.

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The normalized pair $(\alpha, \beta)^T$ is the Jones vector. If the two components have the same phase, the tip of $\mathbf{E}$ moves back and forth along a line. A quarter-cycle relative phase with equal magnitudes produces a circle. General amplitudes and relative phases produce an ellipse.

The examples on the board may be written dimensionally as

$$E_x = E_0 \cos(kz - \omega t), \quad (1.33)$$

$$E_y = E_0 \sin(kz - \omega t), \quad (1.34)$$

which traces a circle at fixed z , and

$$E_x = E_0 \cos(kz - \omega t), \quad (1.35)$$

$$E_y = \frac{E_0}{2} \sin(kz - \omega t), \quad (1.36)$$

which traces an ellipse with its major axis along x .

The ellipse needs two independent real parameters. One gives the orientation of its major axis. The other gives its shape together with the sense of rotation. The lecture informally calls the latter a signed eccentricity. In standard optics it is cleaner to use the *signed ellipticity angle* ϵ :

$$-\frac{\pi}{4} \leq \epsilon \leq \frac{\pi}{4}, \quad |\tan \epsilon| = \frac{\text{minor semiaxis}}{\text{major semiaxis}}. \quad (1.37)$$

The sign records handedness. Linear polarization has $\epsilon = 0$; circular polarization has $|\epsilon| = \pi/4$, where the major-axis direction becomes meaningless.

Question from the audience. Do clockwise and counterclockwise polarization require a third parameter?^a

Response. No. The shape parameter may carry a sign. As an ellipse is squeezed through the linear-polarization limit, changing the sign reverses the sense in which the field traverses the curve. Standard optics expresses this with signed ellipticity rather than literal signed geometric eccentricity.

^aLecture timestamp: 00:37:52–00:39:00.

1.5.1 Reading the Ellipse from Analyzer Probabilities

Let

$$b = \sqrt{1 - a^2}, \quad |\psi\rangle = \begin{pmatrix} a \\ be^{i\phi} \end{pmatrix}, \quad (1.38)$$

and compare it with a linearly polarized analyzer state

$$|\theta\rangle = \begin{pmatrix} \cos \theta \\ \sin \theta \end{pmatrix}. \quad (1.39)$$

The two conjugate amplitudes are

$$\langle \theta | \psi \rangle = a \cos \theta + be^{i\phi} \sin \theta, \quad (1.40)$$

$$\langle \psi | \theta \rangle = a \cos \theta + be^{-i\phi} \sin \theta. \quad (1.41)$$

Question from the audience. Should one of the phase factors in the overlap be complex conjugated?^a

Response. Yes. The sign depends on the bra-ket order. In $\langle \psi | \theta \rangle$, the coefficient from $|\psi\rangle$ is conjugated, giving $e^{-i\phi}$. The opposite order gives $e^{i\phi}$. Their absolute squares agree.

^aLecture timestamp: 00:41:48–00:42:02.

The transmission probability is

$$P_\theta = |\langle \theta | \psi \rangle|^2 \quad (1.42)$$

$$= a^2 \cos^2 \theta + b^2 \sin^2 \theta + 2ab \cos \phi \sin \theta \cos \theta \quad (1.43)$$

$$= \frac{1}{2} \left[1 + (a^2 - b^2) \cos 2\theta + 2ab \cos \phi \sin 2\theta \right]. \quad (1.44)$$

The lecture leaves this multiplication and maximization as homework. Finishing that step makes its geometric prescription exact. The major-axis angle θ_{maj} obeys

$$2\theta_{\text{maj}} = \text{atan2} \left(2ab \cos \phi, a^2 - b^2 \right) \pmod{2\pi}. \quad (1.45)$$

Thus the analyzer most likely to transmit the photon points along the major axis of the polarization ellipse.

It is useful to collect the same information in the normalized Stokes parameters

$$S_1 = a^2 - b^2, \quad S_2 = 2ab \cos \phi, \quad S_3 = 2ab \sin \phi, \quad (1.46)$$

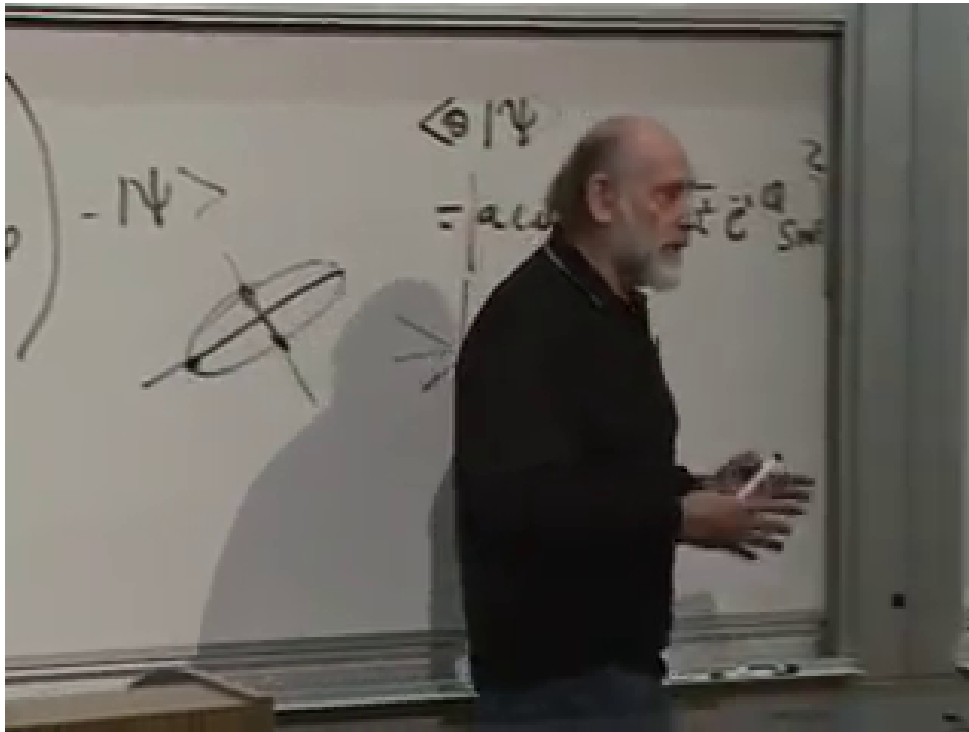


Figure 1.8: The overlap amplitude and the geometric search for the linear-analyzer direction of maximum transmission.

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for which $S_1^2 + S_2^2 + S_3^2 = 1$. With a fixed propagation and time convention,

$$2\theta_{\text{maj}} = \text{atan2}(S_2, S_1), \quad \sin 2\epsilon = S_3. \quad (1.47)$$

Changing the convention for viewing the wave reverses the sign assigned to handedness, but not the two-parameter count.

Question from the audience. Orientation and ordinary eccentricity still seem to omit clockwise versus counterclockwise motion. Where is it stored? ^a

Response. It is stored in the sign of the second parameter. The major-axis angle gives the orientation; signed ellipticity gives both the axis ratio and the traversal sense. At the linear limit the sign can change, and at the circular limit the axis angle is irrelevant.

^aLecture timestamp: 00:45:31–00:47:14.

1.6 Time Evolution Must Preserve Distinctions

1.6.1 The Classical Analogy

We now turn from the description of states to their motion. A finite classical state set admits only jumps if time is viewed microscopically, but the quantum vector space contains continuous paths even when a measurement has only two possible outcomes. A linear polarization axis can rotate continuously with time although every fixed analyzer still records only $+1$ or -1 .

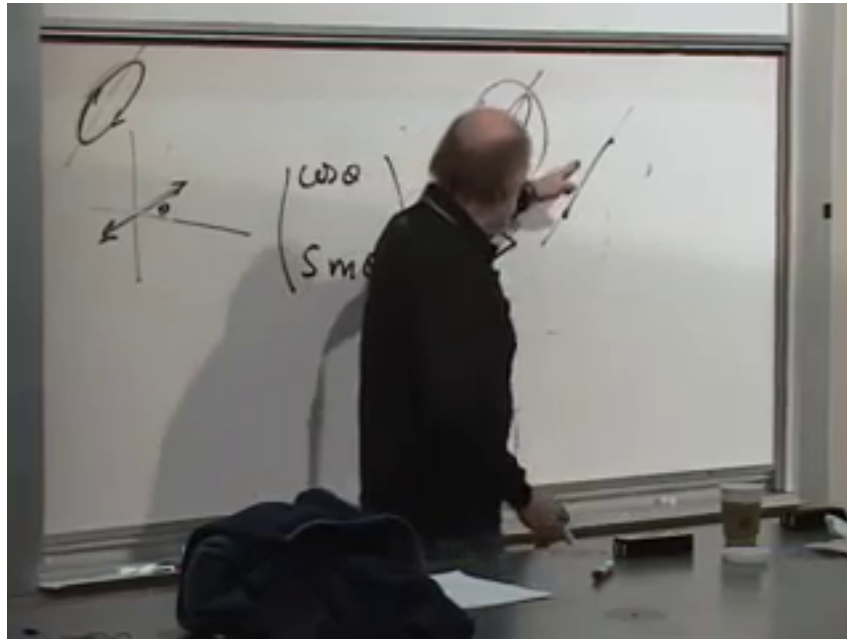


Figure 1.9: Squeezing through the linear limit reverses the traversal sense, which is encoded by the sign of ellipticity.

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For a closed reversible classical system, evolution is one-to-one. It may permute the states, but it cannot send two distinct initial states to the same final state. If both 5 and 2 on a die evolved to 3, the final state would not reveal which initial state had been prepared. The distinction would have been erased.

Quantum states have more structure than simply same or different. For normalized vectors, the invariant transition probability

$$|\langle B | A \rangle|^2 \quad (1.48)$$

measures their operational overlap. Orthogonal states have zero overlap and can be distinguished with certainty. States on the same ray have $|\langle B | A \rangle| = 1$. Intermediate values mean that no single ideal test distinguishes the preparations with certainty.

1.6.2 Preserving the Inner Product

Prepare two identical systems in $|A\rangle$ and $|B\rangle$, and let both evolve for the same time:

$$|A\rangle \longrightarrow |A'\rangle, \quad |B\rangle \longrightarrow |B'\rangle. \quad (1.49)$$

Closed-system quantum evolution preserves their inner product:

$$\boxed{\langle B' | A' \rangle = \langle B | A \rangle}. \quad (1.50)$$

Consequently equal rays remain equal rays, orthogonal states remain orthogonal, and every transition probability remains unchanged. This is the quantum form of preserving distinctions.

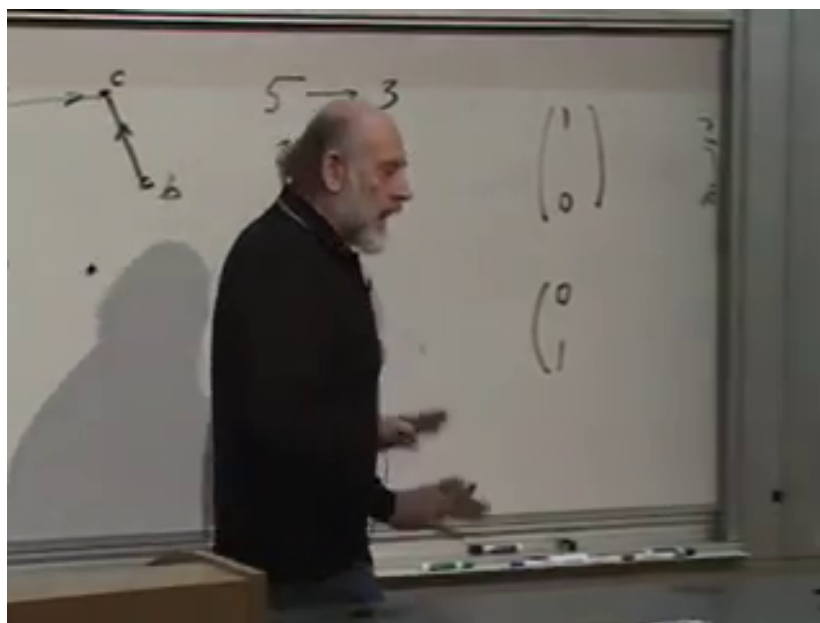


Figure 1.10: A forbidden many-to-one rule would send both 5 and 2 to 3, erasing the initial distinction.

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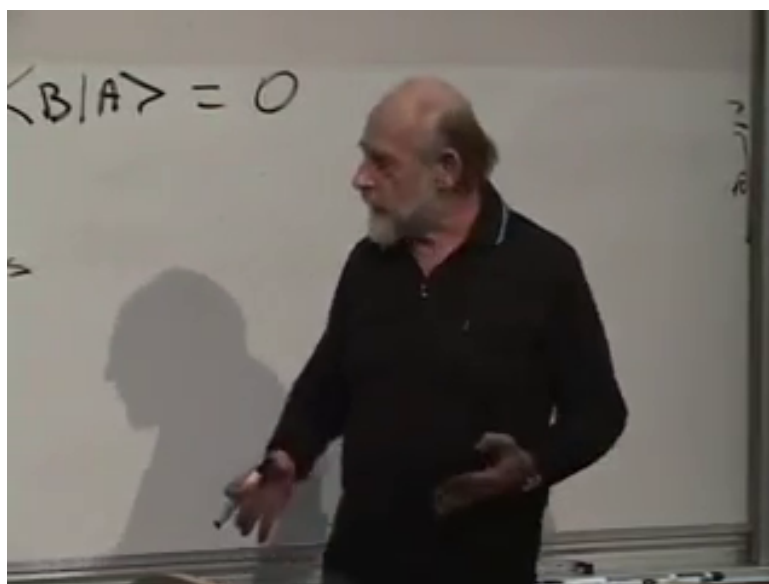


Figure 1.11: Zero overlap represents orthogonal, perfectly distinguishable polarization states.

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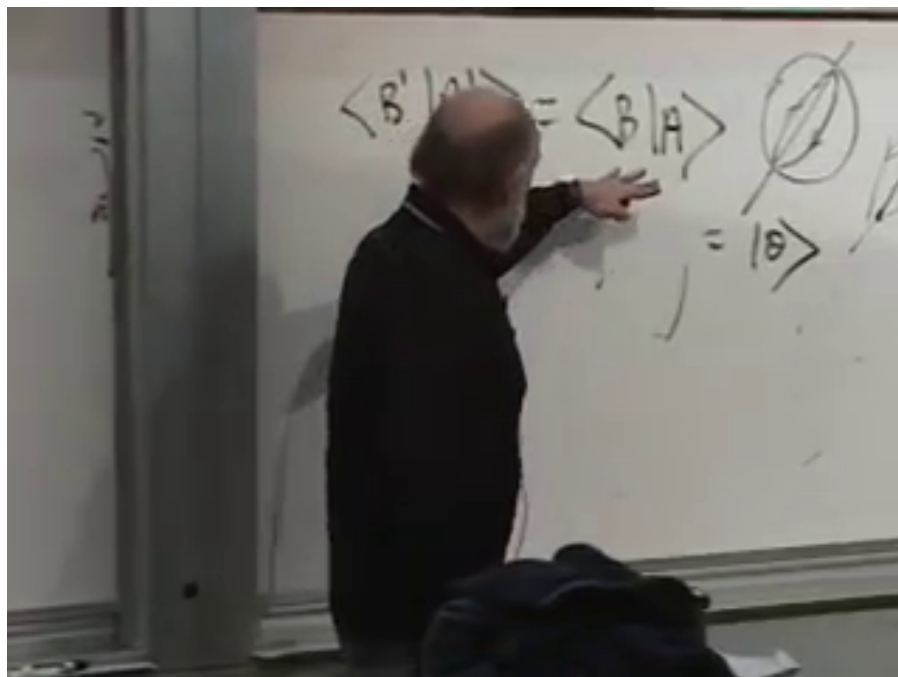


Figure 1.12: The overlap of two states is unchanged when both undergo the same closed-system evolution.

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Question from the audience. If one photon fails a horizontal analyzer, how do we know it was vertical rather than a 45° state that happened to fail? ^a

Response. From one event, we do not. Orthogonality is tested by certainty across identically prepared trials. Every genuinely vertical photon is rejected by an ideal horizontal analyzer; a 45° state is rejected only with probability one half. The ensemble distinguishes the preparations.

^aLecture timestamp: 00:58:17–00:59:00.

Question from the audience. Does the uncertainty principle force a small chance that a perfectly horizontal photon passes a vertical polarizer? ^a

Response. No. One polarization observable can be prepared with arbitrary precision in the ideal theory. The uncertainty concerns incompatible observables: no state has definite outcomes for two genuinely different analyzer axes at once. It does not blur an eigenstate of one chosen axis.

^aLecture timestamp: 00:59:01–01:00:31.

In an ordinary real vector space, a transformation preserving all inner products looks like a rigid rotation or reflection: vectors move, but their lengths and mutual angles do not. The complex quantum version is a unitary transformation.

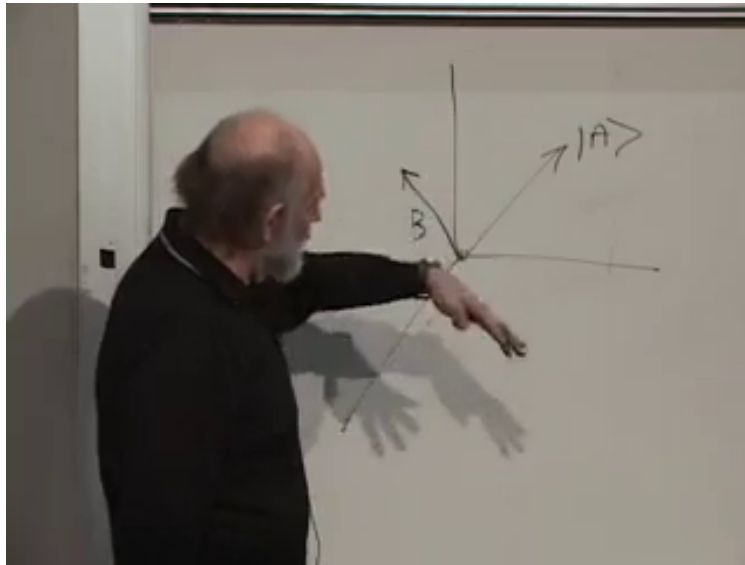


Figure 1.13: The geometric analogy: vectors move while their lengths and mutual angles remain fixed.

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1.7 Operators on Bras and the Adjoint

Let $\hat{L}$ be a linear operator on kets:

$$\hat{L} |A\rangle = |C\rangle. \quad (1.51)$$

Linearity means

$$\hat{L}(\alpha |A\rangle + \beta |B\rangle) = \alpha \hat{L} |A\rangle + \beta \hat{L} |B\rangle. \quad (1.52)$$

For any bra $\langle B|$, the matrix element

$$\langle B| \hat{L} |A\rangle \quad (1.53)$$

may be read associatively as the inner product of $\langle B|$ with $\hat{L} |A\rangle$, or as the bra $\langle B| \hat{L}$ applied to $|A\rangle$.

Inserting a complete orthonormal basis makes the latter bra explicit:

$$\langle B| \hat{L} = \sum_n \langle B| \hat{L} |n\rangle \langle n|. \quad (1.54)$$

The coefficients are numbers, so the right side is a linear combination of bras.

Question from the audience. Does this construction assume that $\hat{L}$ is Hermitian?^a

Response. No. The matrix element and the bra $\langle B| \hat{L}$ are defined for a general linear operator. Hermiticity is the additional condition $\hat{L}^\dagger = \hat{L}$, introduced later.

^aLecture timestamp: 01:05:51–01:06:00.

Question from the audience. Are we assuming associativity when the brackets are re-grouped?^a

Response. Yes. Operator composition and its action on vectors are associative wherever the products are defined: $\langle B| (\hat{L} |A\rangle) = (\langle B| \hat{L}) |A\rangle$. This is what lets the notation be manipulated consistently.

^aLecture timestamp: 01:06:01–01:06:18.

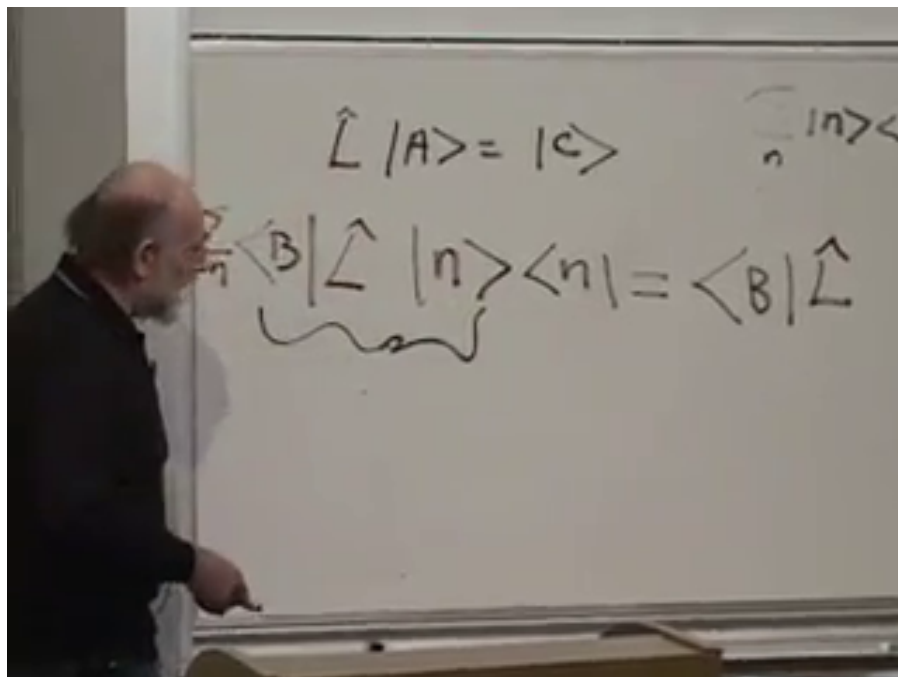


Figure 1.14: A completeness insertion defines the bra $\langle B | \hat{L}$ from the known action of $\hat{L}$ on kets.

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1.7.1 Hermitian Conjugation

Taking the adjoint of Equation (1.51) gives

$$\langle A | \hat{L}^\dagger = \langle C |. \quad (1.55)$$

The defining matrix-element identity is

$$\boxed{\langle B | \hat{L} | A \rangle = \left(\langle A | \hat{L}^\dagger | B \rangle \right)^*}. \quad (1.56)$$

It combines two operations already familiar from inner products: interchange bra and ket, then complex conjugate.

In an orthonormal basis, define

$$L_{mn} = \langle m | \hat{L} | n \rangle. \quad (1.57)$$

Then

$$(\hat{L}^\dagger)_{mn} = L_{nm}^*. \quad (1.58)$$

Thus the matrix of $\hat{L}^\dagger$ is the complex-conjugate transpose:

$$L^\dagger = (L^*)^T. \quad (1.59)$$

Rows and columns are interchanged, and every entry is conjugated.

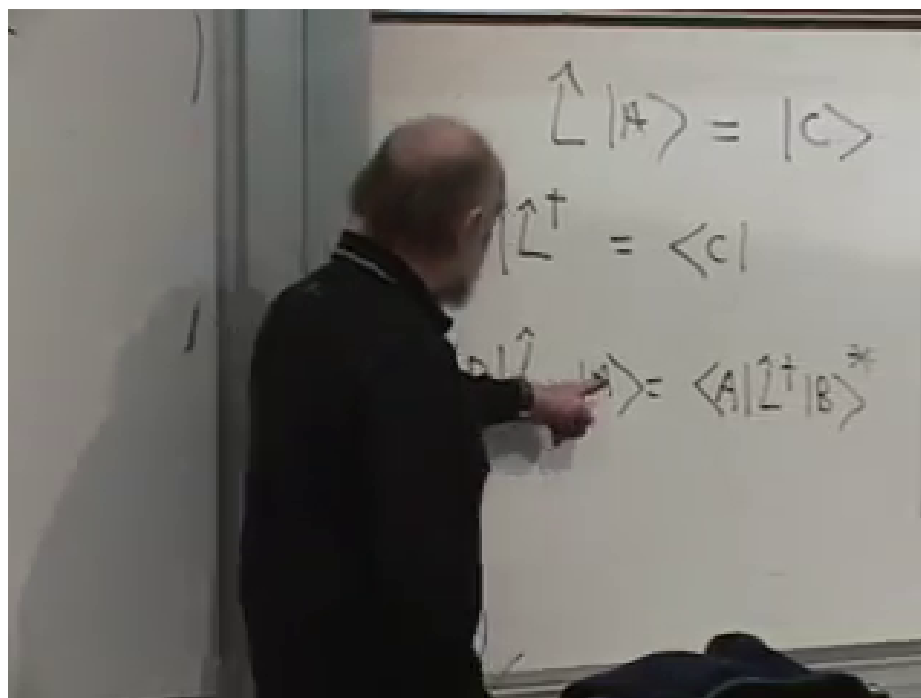


Figure 1.15: The adjoint relation obtained by reversing the bra-ket order and complex conjugating the matrix element.

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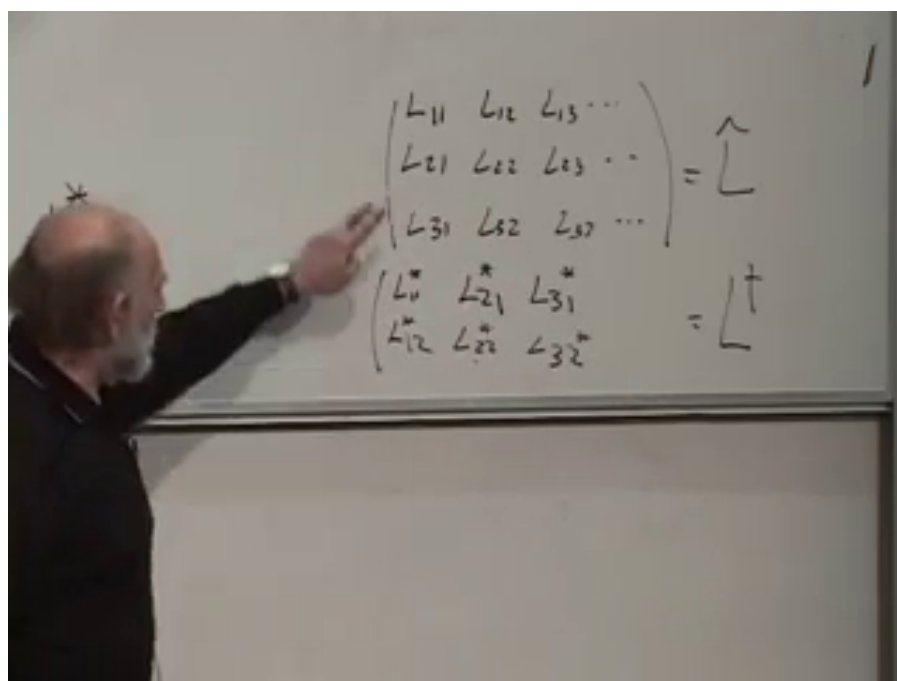


Figure 1.16: The matrix rule for the adjoint: transpose and complex conjugate.

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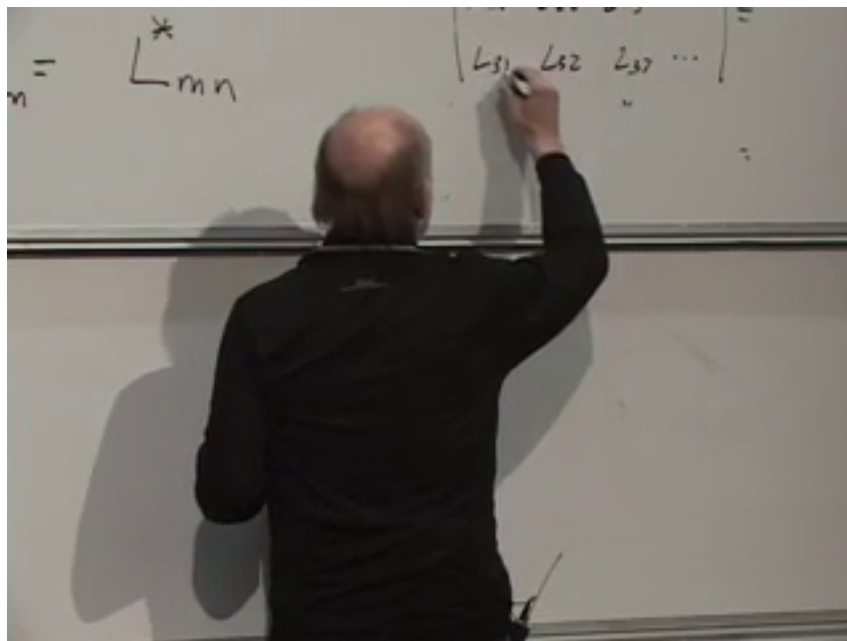


Figure 1.17: The conjugate-symmetry condition $L_{mn} = L_{nm}^*$ for a Hermitian matrix.

Lecture frame: 01:19:30.

Question from the audience. Are we assuming a finite-dimensional state space here? ^a

Response. For the displayed matrices, yes. The same adjoint identity extends to Hilbert spaces, but infinite-dimensional operators require care about domains and completeness. The finite-dimensional case contains the algebra needed for the present argument without those analytic qualifications.

^aLecture timestamp: 01:13:44–01:14:05.

1.8 Hermitian Operators Represent Observables

An operator is Hermitian when

$$\hat{L} = \hat{L}^\dagger. \quad (1.60)$$

In components,

$$L_{mn} = L_{nm}^*. \quad (1.61)$$

The diagonal entries are therefore real,

$$L_{nn} = L_{nn}^* \in \mathbb{R}, \quad (1.62)$$

while entries reflected across the diagonal are complex conjugates.

In finite dimensions, the spectral theorem gives the properties needed of an observable:

1. every eigenvalue is real;
2. eigenvectors belonging to distinct eigenvalues are orthogonal;
3. an orthonormal eigenbasis exists.

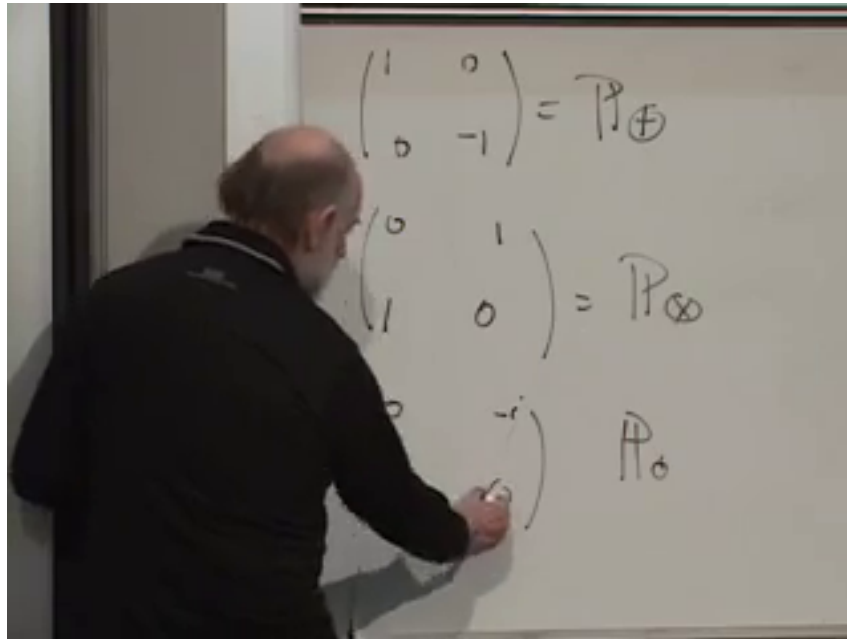


Figure 1.18: The axial, diagonal, and circular polarization observables are all Hermitian.

Lecture frame: 01:24:45.

The eigenvectors need not have real components. Circular polarization is the immediate counterexample: the Hermitian matrix $\hat{P}_c$ has eigenvectors containing $\pm i$.

The expectation value of a Hermitian operator is real:

$$\left(\langle \psi | \hat{L} | \psi \rangle \right)^* = \langle \psi | \hat{L}^\dagger | \psi \rangle \quad (1.63)$$

$$= \langle \psi | \hat{L} | \psi \rangle. \quad (1.64)$$

Conversely, in finite dimensions, if $\langle \psi | \hat{L} | \psi \rangle$ is real for every $|\psi\rangle$, then $\hat{L}$ is Hermitian.

Question from the audience. How large is the matrix representing an observable? ^a

Response. It matches the dimension of the state space under study. Polarization alone is two-dimensional. A full photon state also carries continuous momentum or position labels and is generally infinite-dimensional. Finite truncations are often used for calculation, but finite systems such as an isolated spin degree of freedom are not merely approximations.

^aLecture timestamp: 01:20:28–01:22:28.

The three polarization observables make a compact check:

$$\hat{P}_0 = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}, \quad \hat{P}_{\pi/4} = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad \hat{P}_c = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}. \quad (1.65)$$

Each equals its adjoint. They are, respectively, the Pauli matrices $\sigma_z, \sigma_x, \sigma_y$.

Question from the audience. Why is Hermiticity the right condition for an observable? ^a

Response. Laboratory outcomes and their averages are real numbers. Hermiticity guarantees real eigenvalues and real expectation values. In finite dimensions the converse condition on all quadratic forms also recovers Hermiticity, so the mathematical requirement is tightly matched to the operational one.

^aLecture timestamp: 01:25:08–01:26:20.

Question from the audience. Must the eigenvectors of an observable span the whole state space? ^a

Response. Yes in the finite-dimensional Hermitian case: the spectral theorem supplies a complete orthonormal eigenbasis, including an orthonormal choice inside degenerate eigenspaces. A general non-Hermitian operator may have nonorthogonal eigenvectors or may fail to have a complete eigenbasis.

^aLecture timestamp: 01:26:22–01:27:43.

1.9 Unitary Operators Preserve Quantum Relations

Hermitian operators represent quantities that can be measured. They are not, in general, the transformations that move states while preserving their relations. Let one linear operator $\hat{U}$ act on every state:

$$\hat{U} |A\rangle = |A'\rangle, \quad \hat{U} |B\rangle = |B'\rangle. \quad (1.66)$$

The transformed bras are

$$\langle A'| = \langle A| \hat{U}^\dagger, \quad \langle B'| = \langle B| \hat{U}^\dagger. \quad (1.67)$$

Therefore

$$\langle B' | A' \rangle = \langle B | \hat{U}^\dagger \hat{U} | A \rangle. \quad (1.68)$$

Demanding equality with $\langle B | A \rangle$ for every pair $|A\rangle, |B\rangle$ forces

$$\boxed{\hat{U}^\dagger \hat{U} = \mathbf{1}.} \quad (1.69)$$

In finite dimensions this condition makes $\hat{U}$ invertible and implies

$$\hat{U}^\dagger = \hat{U}^{-1}, \quad \hat{U} \hat{U}^\dagger = \hat{U}^\dagger \hat{U} = \mathbf{1}. \quad (1.70)$$

In an infinite-dimensional Hilbert space, the definition of a unitary operator includes both equalities; $\hat{U}^\dagger \hat{U} = \mathbf{1}$ alone defines an isometry that need not be onto.

A unitary transformation is not the identity. It may rotate polarization, implement a symmetry, or carry a state from one time to another. Its adjoint undoes it, and throughout the motion

$$\langle UB | UA \rangle = \langle B | \hat{U}^\dagger \hat{U} | A \rangle = \langle B | A \rangle. \quad (1.71)$$

Thus every norm, orthogonality relation, and transition probability is preserved.

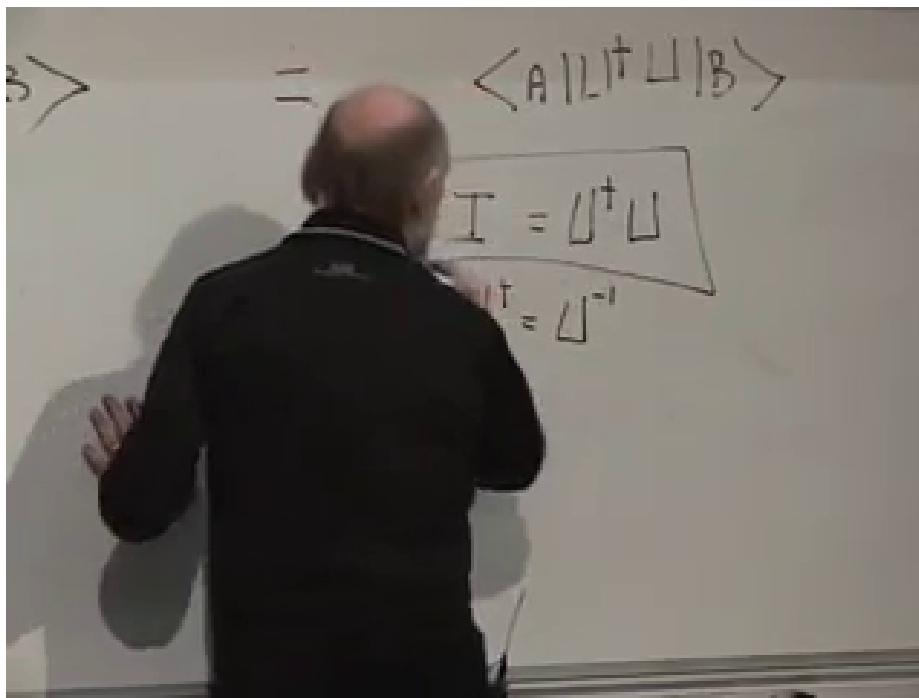


Figure 1.19: Preservation of every inner product leads to the unitary condition $\hat{U}^\dagger \hat{U} = \mathbf{1}$.

Lecture frame: 01:34:50.

Question from the audience. Does preserving relationships mean anything broader than preserving inner products?^a

Response. Mathematically, inner-product preservation is the precise condition used here. Physically it preserves the absolute squares of those inner products and therefore every transition probability between simultaneously evolved preparations.

^aLecture timestamp: 01:35:13–01:35:53.

The lecture closes just as the eigenvalue structure of a unitary operator is introduced:

$$\hat{U} |n\rangle = \lambda_n |n\rangle. \quad (1.72)$$

The unanswered question has a short consequence of unitarity:

$$1 = \langle n | n \rangle = \langle n | \hat{U}^\dagger \hat{U} | n \rangle \quad (1.73)$$

$$= |\lambda_n|^2 \langle n | n \rangle, \quad (1.74)$$

so

$$|\lambda_n| = 1, \quad \lambda_n = e^{i\omega_n}. \quad (1.75)$$

Finite-dimensional unitary matrices are normal and therefore possess a complete orthonormal eigenbasis, but their eigenvalues are phases rather than necessarily real numbers.

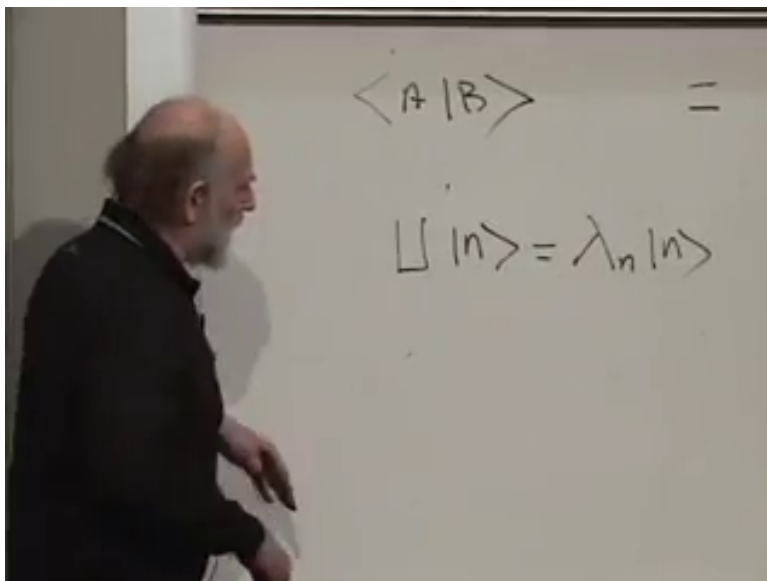


Figure 1.20: The final board question: the eigenvalues of a unitary operator lie on the unit circle.

Lecture frame: 01:37:20.

We have reached the threshold of dynamics. Closed systems evolve by unitary transformations. The next step is to discover how continuous unitary evolution is generated by a Hermitian Hamiltonian and how that statement becomes the Schrodinger equation.